

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 95-263

WASTE DISCHARGE REQUIREMENTS
FOR
FRESNO COUNTY SERVICE AREA NO. 31B
WASTEWATER TREATMENT FACILITY
FRESNO COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. Fresno County Service Area No. 31B (Discharger) owns and operates the County Service Area (CSA) No. 31B Wastewater Treatment Facility (WWTF), which serves the unincorporated community of Shaver Lake Village.
2. Waste Discharge Requirements Order No. 80-190, adopted by the Board on 5 December 1980, prescribes requirements for a discharge of 1.0 million gallons per day (mgd) of treated domestic wastewater from a proposed expanded and improved wastewater treatment facility at County Water Works District (WWD) No. 41, which was supposed to provide sewerage services to areas within both CSA No. 31B and WWD No. 41.
3. The facility at WWD No. 41 was never expanded. Instead, the Discharger built the CSA No. 31B WWTF for Shaver Lake Village in 1986 with assistance in the form of a Clean Water Grant.
4. The WWTF at CSA No. 31B has a design treatment capacity of 0.230 mgd and consists of an Imhoff tank, an anaerobic sludge digester, two sludge drying beds, and seven percolation/evaporation ponds. Dried sludge from the WWTF is disposed of at one of County's permitted landfills.
5. Wastewater from Shaver Lake Village and the housing south of it converge in a trunk line, which ends at a flow splitter box. County staff apportions wastewater between the CSA No. 31B WWTF and the WWD No. 41 WWTF. Monthly flows into CSA No. 31B WWTF average about 0.064 mgd. The average flow for July 1995 was 0.168 mgd.
6. The WWTF is in Section 3, T10S, R24E, MDB&M, with surface water drainage to Falls Creek, as shown in Attachment A, which is attached hereto and part of this Order by reference. The site lies within the Redinger hydrologic area (No. 540.30), as depicted on interagency hydrologic maps prepared by the Department of Water Resources in August 1986.
7. The Board adopted a Water Quality Control Plan, Third Edition, for the Sacramento River and San Joaquin River Basins (hereafter Basin Plan), which contains water quality objectives for waters of the Basin. These requirements implement the Basin Plan.

WASTE DISCHARGE REQUIREMENTS
FRESNO COUNTY SERVICE AREA 31B WWTF
FRESNO COUNTY

-2-

8. Falls Creek is tributary to the San Joaquin River, a water of the United States. Several ephemeral streams tributary to the creek cut across the disposal area.
9. The beneficial uses of this reach of the San Joaquin River are municipal, industrial, and agricultural supply; recreation; esthetic enjoyment; hydropower generation; and preservation and enhancement of fish, wildlife, and other aquatic resources.
10. The beneficial uses of underlying ground water are domestic, industrial, and agricultural supply.
11. The WWTF is on a steep hillside, and includes seven evaporation/percolation disposal ponds which are cut into the hillside. Disposal Pond No. 3 was built on top of a natural spring and for that reason it has never been used for disposal of wastes. Soils beneath pond bottoms consist of weathered granite underlain by granitic bedrock cut by joints and fractures. Pond percolation rates are between 0.1 and 0.13 feet/day.
12. Board inspection records show that precipitation during winter months renders the sludge drying beds useless and prevents the Discharger from wasting sludge from the Imhoff tank, which in turn interferes with proper WWTF operation and maintenance. If the sludge is not removed from the beds before the winter, the beds basically turn into lagoons, which threaten overflows into the surrounding ephemeral surface water streams, surface water drainage channels, and natural springs.
13. Areal ground water is four to eight feet below ground surface and of good to excellent quality. On-site discharges are undisinfected. There is very little soil structure beneath pond bottoms to remove bacteria concentrations that threaten ground water quality. The Discharger installed one upgradient and three downgradient wells for the disposal ponds. The upgradient well is dry. Ground water monitoring results show that nitrate concentrations in ground water downgradient of the discharge have not exceeded the 45 mg/l State MCL, but are increasing steadily. The results also shows that ground water downgradient of the discharge is polluted water with bacteria.
14. The Discharger is finalizing a wastewater master plan for the Shaver Lake area. The master plan establishes approximate size, location, and costs for future wastewater treatment and disposal facilities for CSA No. 31 and WWD No. 41. According to the master plan, the facilities would be incrementally expanded to ultimately treat and dispose of wastewater generated by up to 3647 equivalent dwelling units (i.e., 1.0 mgd). The proposed expansions are not expected to materialize in the immediate future and are not addressed in this Order.

WASTE DISCHARGE REQUIREMENTS
FRESNO COUNTY SERVICE AREA 31B WWTF
FRESNO COUNTY

-3-

15. The Discharger certified a final environmental impact report (EIR) on 7 August 1980 in accordance with the California Environmental Quality Act (CEQA) (Public Resources Code Section 21000, et seq.) and the State CEQA Guidelines for the proposed interim expansion.
16. Compliance with these waste discharge requirements will mitigate or avoid the potential adverse water quality and resources impacts identified in Finding No. 13, above.
17. The permitted discharge is consistent with the antidegradation provisions of State Water Resources Control Board Resolution 68-16. This Order provides for no increase in the volume of pollutants discharged from the existing facility. The project's discharge to the disposal area will not cause significant impacts on ground water and surface waters, or depletion of limited ground water resources. The discharge allows wastewater utility service necessary for the protection of the public and economic expansion in the area, and is considered to be a benefit to the people of the State.
18. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
19. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Fresno County Service Area No. 31B, its agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

A. Discharge Prohibitions:

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated waste is prohibited.
3. Discharge of collected screenings, sludges, and other solids removed from liquid wastes to uncovered surface impoundments, ponds, and sludge drying beds from 1 November and 30 May of each year is prohibited.
4. Discharge of waste classified as 'hazardous' or 'designated', as defined in Sections 2521(a) and 2522(a) of Title 23, California Code of Regulations, Section 2510, et seq., (hereafter Chapter 15), is prohibited.

WASTE DISCHARGE REQUIREMENTS
FRESNO COUNTY SERVICE AREA 31B WWTF
FRESNO COUNTY

-4-

B. Discharge Specifications:

1. The monthly average dry weather discharge shall not exceed 0.230 mgd.
2. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater treatment and disposal areas.
3. As a means of discerning compliance with Discharge Specification No. B.2, the dissolved oxygen content in the upper zone (1 foot) of wastewater in ponds shall not be less than 1.0 mg/l.
4. The effluent from the treatment facility shall not exceed the following limits:

<u>Constituent</u>	<u>Units</u>	<u>Monthly Average</u>	<u>Daily Maximum</u>
BOD ₅ ¹	mg/l	40	80
Total Coliform Organisms	MPN/100 ml	23	240
Settleable Solids	ml/l	0.2	0.5

¹ 5-day, 20° Celsius biochemical oxygen demand.

5. Ponds shall not have a pH less than 6.5 or greater than 9.5
6. Ponds shall be managed to prevent breeding of mosquitos. In particular:
 - a. An erosion control program should assure that small coves and irregularities are not created around the perimeter of the water surface.
 - b. Weeds shall be minimized through control of water depth, harvesting, or herbicides.
 - c. Dead algae, vegetation, and debris shall not accumulate on the water surface.
7. Public contact with wastewater in the treatment and disposal areas shall be precluded through such means as fences, signs, and other acceptable alternatives.

WASTE DISCHARGE REQUIREMENTS
FRESNO COUNTY SERVICE AREA 31B WWTF
FRESNO COUNTY

-5-

8. Ponds shall have sufficient capacity to accommodate allowable wastewater flow and design seasonal precipitation and ancillary inflow and infiltration. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years, distributed monthly in accordance with historical rainfall patterns. Freeboard shall never be less than two feet (measured vertically).
9. On or about 1 October of each year, available pond storage capacity shall at least equal the volume necessary to comply with Discharge Specification No. B.8.

C. Sludge Disposal Specifications:

1. Collected screenings, sludges, and other solids removed from liquid wastes shall be removed from the sludge drying beds on or before 1 November of each year and disposed of in a manner that is consistent with Chapter 15 and approved by the Executive Officer.
2. Any proposed change in sludge use or disposal practice shall be reported to the Executive Officer **at least 90 days in advance** of the change.
3. Use and disposal of sewage sludge shall comply with existing federal and state laws and regulations, including permitting requirements and technical standards included in 40 CFR 503.

If the State Water Resources Control Board and the regional water quality control boards accept primacy to implement regulations contained in 40 CFR 503, this Order may be reopened to incorporate appropriate time schedules and technical standards. The Discharger must comply with the standards and time schedules contained in 40 CFR 503 whether or not they have been incorporated into this Order.

D. Ground Water Limitations:

The discharge, in combination with other sources, shall not cause underlying ground water to contain waste constituents in concentrations statistically greater than background water quality.

E. Provisions:

1. The Discharger shall comply with Monitoring and Reporting Program No. 95-263, which is part of this Order, and any revisions thereto as ordered by the Executive Officer.

WASTE DISCHARGE REQUIREMENTS
FRESNO COUNTY SERVICE AREA 31B WWTF
FRESNO COUNTY

-6-

2. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements," dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."
3. The Discharger shall comply with all the terms and conditions of this Order shall be achieved according to the following time schedule:
 - a. **By 10 January 1996**, the Discharger shall submit a work plan to develop a comprehensive ground water monitoring network to monitor the effects of the discharge on ground water quality, determine ground water gradient and direction of flow of ground water within the influence of the discharge. The proposed network shall include, but not be limited to: (1) a proposed new background well, (2) a map showing the location of existing and proposed wells, and (3) the construction features of the wells. The report shall contain a time schedule for implementation by **15 March 1996**.
 - b. **By 15 September 1996**, the Discharger shall submit a technical report consisting of plans and specifications for compliance with effluent limits and disinfection requirements of Discharge Specification No. B.4. The report shall contain a time schedule for implementation by **1 November 1996**; and
 - c. **By 30 October 1996**, the Discharger shall provide a technical report for sludge storage and disposal for the WWTF. The report shall: (1) address compliance with Sludge Disposal Specification Nos. C.1 to C.3; (2) describe the proposed measures to store and dispose of collected screenings, sludges, and other solids removed from liquid wastes from 1 November to 30 May of each year; (3) include a time schedule for implementation of the proposed measures; and (4) certify that the proposed measures do not interfere with the proper operation and maintenance of the WWTF;

All technical reports must be prepared under the direction of a California Registered Civil Engineer with experience in the design and operation of domestic WWTFs. All reports are subject to the review and approval of the Executive Officer.

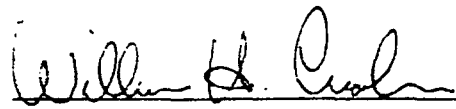
4. The Discharger shall use the best practicable, cost-effective control technique currently available to comply with limits specified in this order.

WASTE DISCHARGE REQUIREMENTS
FRESNO COUNTY SERVICE AREA 31B WWTF
FRESNO COUNTY

-7-

5. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.
6. The Discharger shall submit to the Board on or before each report due date the specified document or, if an action is specified, a written report detailing evidence of compliance with the date and task. If noncompliance is being reported, the reasons for such noncompliance shall be stated, plus an estimate of the date when the Discharger will be in compliance. The Discharger shall notify the Board by letter when it returns to compliance with the time schedule.
7. A copy of this Order shall be kept at the WWTF for reference by operating personnel. Key operating personnel shall be familiar with its contents.
8. If reclaimed water is used for construction purposes, it shall comply with the most current edition of "Guidelines for Use of Reclaimed Water for Construction Purposes". Other uses of reclaimed water not specifically authorized herein shall be subject to the approval of the Executive Officer and shall comply with 22 CCR, Division 4.
9. The Board will review this Order periodically and will revise requirements when necessary.

I, WILLIAM H. CROOKS, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 8 December 1995.


WILLIAM H. CROOKS, Executive Officer

JA:ja/fmc

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 95-263

FOR

FRESNO COUNTY SERVICE AREA NO. 31B

WASTEWATER TREATMENT FACILITY

FRESNO COUNTY

EFFLUENT MONITORING

Except for flow, which may also be measured at the headworks, effluent samples shall be collected prior to discharge to the disposal ponds. Effluent samples should be representative of the volume and nature of the discharge. Time of collection of a grab sample shall be recorded. Effluent monitoring shall include at least the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Frequency</u>
Total Daily Flow	gpd	Continuous	Daily
Settleable Solids	ml/l	Grab	Weekdays
BOD ₅ ²	mg/l	Grab	Weekly
Total Coliform Organisms	MPN/100 ml	Grab	Weekly

¹ If results of monitoring a pollutant appear to violate effluent limitations, but monitoring frequency is not sufficient to validate violation (e.g., the monthly average for BOD), or indicate a violation and potential upset of the treatment process (e.g., less than minimum D.O.), the frequency of sampling shall be increased to confirm the magnitude and duration of violation, if any, and aid in identification and resolution of the problem.

² Five-day, 20° Celsius biochemical oxygen demand.

POND MONITORING

The freeboard shall be monitored on the storage pond to the nearest tenth foot. Pond water monitoring shall include the following :

<u>Constituent</u>	<u>Unit</u>	<u>Measurement</u>	<u>Frequency</u>
Freeboard	feet	Observation	Weekly
pH	pH units	Grab	Daily
Dissolved Oxygen ¹	mg/l	Grab	Weekdays

¹ Samples shall be collected at a depth of one foot from each pond, opposite the inlet, and analyzed for dissolved oxygen. Samples shall be collected between 0800 and 0900 hours.

MONITORING AND REPORTING PROGRAM
FRESNO COUNTY SERVICE AREA NO. 31B WWTF
FRESNO COUNTY

-2-

Permanent markers (e.g., staff gages) shall be placed in the pond with calibration indicating the water level at design capacity and available operational freeboard. In addition, the Discharger shall inspect the conditions of the ponds once per week and write visual observation in a bound log book. Notations shall include observations of whether weeds are developing in the water or along the bank, and their locations; whether dead algae, vegetation, scum, or debris are accumulating on the pond surface and their location; whether burrowing animals or insects are present; and the color of the pond (e.g., dark sparking green, dull green, yellow, grey, tan, brown, etc.) A copy of the entries made in the log during each month shall be submitted along with the monitoring report the following month. Where the O&M manual indicates remedial action is necessary, the Discharger shall briefly explain in the transmittal what action has been taken or is scheduled to be taken.

GROUND WATER MONITORING

Pursuant to Provision C.4, the Discharger shall sample the approved ground water monitoring wells monthly for one year, and analyze them for the parameters specified below. After one year of sampling the wells, data from these analyses shall be reported to the Board **by the 20th day of the following month** for use in establishing water quality protection standards. If subsequent sampling of background monitoring wells indicates significant water quality changes due to either seasonal fluctuations or other reasons unrelated to waste disposal activities, the Discharger may request modification of the water quality protection standards.

The downgradient wells near the facility property boundaries shall constitute "points of compliance (POCs)". In conjunction with background monitoring, monitoring of POCs will enable one to determine compliance with water quality protection standards. The ground water surface elevation (in feet and hundredths, M.S.L.) in all wells shall be measured on a quarterly basis and used to determine the gradient and direction of ground water flow. Water samples shall be collected from the wells and analyzed as follows:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Specific Conductivity @ 25°C	µmhos/cm	Grab	Quarterly
Chloride	mg/l	Grab	Quarterly
Nitrate Nitrogen	mg/l	Grab	Quarterly
Total Coliform Bacteria	MPN/100 ml	Grab	Quarterly
Depth (bgs) and Elevation (MSL) of Ground Water	Feet & hundredths	Measurement	Quarterly

MONITORING AND REPORTING PROGRAM
FRESNO COUNTY SERVICE AREA NO. 31B WWTF
FRESNO COUNTY

-3-

Following each sampling event, the discharger shall determine whether there is a statistically significant increase over water quality protection standards for each parameter and constituent analyzed. If the Discharger or the Board finds there is a statistically significant increase in indicator parameters or waste constituents over the water quality protection standards at the POC's, the Discharger shall notify the Board immediately and increase the monitoring frequency to monthly, pending additional directions from Board staff.

REPORTING

Monthly monitoring containing results of all sampling for the month shall be submitted to the Board by the **20th day of the following month**.

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in such a manner that illustrates clearly whether the Discharger complies with waste discharge requirements, including calculation of all averages, etc.

If the Discharger monitors any pollutant at the locations designated herein more frequently that is required by this Order, the results of such monitoring shall be included in the discharge monitoring report.

The Discharger may also be requested to submit an annual report to the Board with tabular and graphical summaries of the monitoring data obtained during the previous year. Any such request shall be made in writing. The report shall discuss the corrective actions taken and planned to bring the discharge into full compliance with the waste discharge requirements.

By **31 January of each year**, the Discharger shall submit a written report to the Executive Officer containing the following:

- a. The names, titles, certificate grade and general responsibilities of persons operating and maintaining the wastewater treatment plant.
- b. The names and telephone numbers of persons to contact regarding the plant for emergency and routine situations.
- c. A certified statement of when the flow meter and other monitoring instruments and devices were last calibrated, including identification of who did the calibration (Standard Provision C.4).

MONITORING AND REPORTING PROGRAM
FRESNO COUNTY SERVICE AREA NO. 31B WWTF
FRESNO COUNTY

-4-

- d. A statement whether the current operation and maintenance manual, and contingency plan, reflect the wastewater treatment plant as currently constructed and operated, and the dates when these documents were last reviewed for adequacy.
- e. The total quantity of sludge disposed of during the previous year and ultimate disposal site(s).

All reports submitted in response to this Order shall comply with the signatory requirements in Standard Provision B.3.

The Discharger shall implement the above monitoring program on the first day of the month following effective date of this Order.

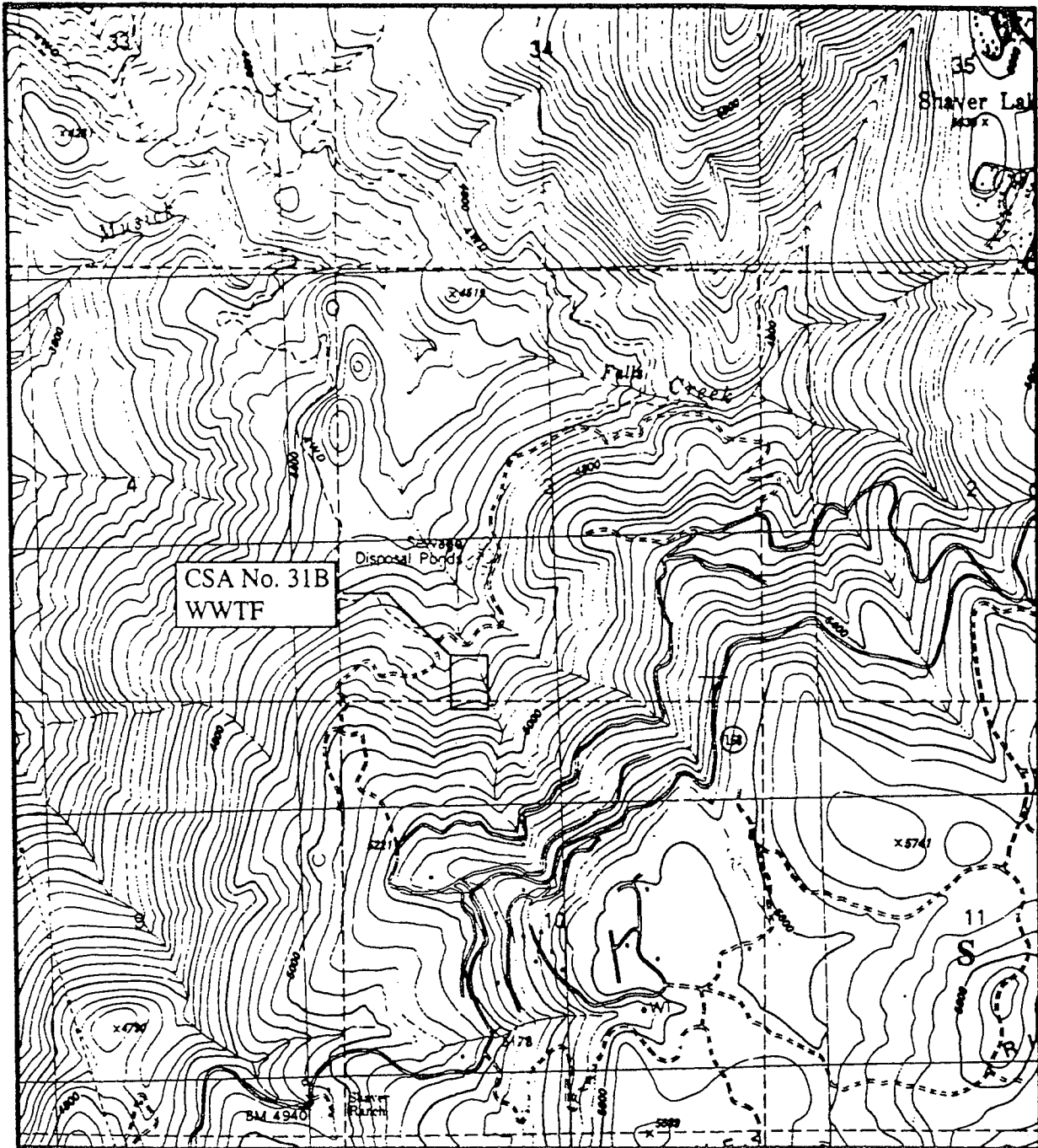
Ordered by:

William H. Crooks
WILLIAM H. CROOKS, Executive Officer

8 December 1995

(Date)

JA:ja/fmc



Scale: 1" = 2000'

ATTACHMENT A

Vicinity Map

Fresno County Service Area No. 31B
Wastewater Treatment Facility
Fresno County

Section 3, T10S, R24E, MDB&M
Shaver Lake, 7 1/2' USGS Quad, 1982

INFORMATION SHEET

FRESNO COUNTY SERVICE AREA NO. 31B WASTEWATER TREATMENT FACILITY FRESNO COUNTY

Fresno County County Service Area (CSA) No. 31 (Discharger) owns and operates its Wastewater Treatment Facility (WWTF), which services the Shaver Lake Village area. In 1980, the Board adopted Waste Discharge Requirements Order No. 80-190 for the discharge of 1.0 million gallons per day (mgd) from a proposed expanded and improved WWTF at Water Works District (WWD) No. 41, which was supposed to provide sewerage services to both CSA No. 31B and WWD No. 41.

The proposed expansion and improvements never materialized. Instead, the Discharger built a 0.230-million gallons per day WWTF for CSA No. 31B southwest of the Shaver Lake Village area. The WWTF consists of an Imhoff tank, an anaerobic sludge digester, two sludge drying beds, and seven percolation/evaporation ponds. Dried sludge from the WWTF is disposed of at one of County's permitted landfills. Wastewater from Shaver Lake Village and the housing south of it converge in a trunk line, which ends at a flow splitter box. The flow splitter apportions wastewater between the WWD No. 41 WWTF and the CSA No. 31B WWTF. The Discharger reports that it uses the facility at WWD No. 41 only during periods of peak flows, like weekends.

The WWTF is on a steep hillside, and includes seven evaporation/percolation disposal ponds which are cut into the hillside. Disposal Pond No. 3 was built on top of a natural spring. Soils beneath pond bottoms consist of weathered granite underlain by granitic bedrock cut by joints and fractures. Pond percolation rates are between 0.1 and 0.13 feet/day. Precipitation renders the sludge drying beds useless during the winter and prevents the Discharger from wasting sludge from the Imhoff tank, which in turn interferes with proper WWTF operation and maintenance. If the sludge is not removed from the beds before the winter, the beds basically turn into lagoons, which threaten overflows into the surrounding ephemeral surface water streams, surface water drainage channels, and natural springs. Areal ground water is four to eight feet below ground surface and of good to excellent quality. On-site discharges are undisinfected. There is very little soil structure beneath pond bottoms to remove bacteria concentrations that threaten ground water quality. The Discharger installed and monitors one upgradient and three downgradient wells for the disposal ponds. The background well is dry and must be replaced. Ground water monitoring results show that nitrate concentration in ground water downgradient of the discharge have not exceeded the 45 mg/l State MCL, but are increasing steadily. The results also shows that ground water downgradient of the discharge is polluted water with bacteria.

This proposed Order requires the Discharger to disinfect the effluent prior to discharge to the ponds. It also requires the Discharger to replace the background ground water monitoring well and develop a comprehensive sludge management plan.

INFORMATION SHEET

FRESNO COUNTY SERVICE AREA NO. 31B
WASTEWATER TREATMENT FACILITY
FRESNO COUNTY

-2-

Fresno County is finalizing a wastewater master plan for the Shaver Lake area. The master plan establishes approximate size, location, and costs for future wastewater treatment and disposal facilities for CSA No. 31 and WWD No. 41. The proposed expansions are not expected to materialize in the immediate future and are not addressed in this Order.

The Discharger certified a final environmental impact report (EIR) in accordance with the California Environmental Quality Act (CEQA) (Public Resources Code Section 21000, et seq.) and the State CEQA Guidelines for the proposed interim expansion.

JA:ja/fmc